Basic Principles of Medicine 2
ER Lecture 13

Male Reproductive Physiology

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Recommended reading

- Rhoades RA, Bell DR. Medical Physiology: Principles for Clinical Medicine, 6e Chapter 36, Pages 773-790. Male Reproductive System | Medical Physiology: Principles for Clinical Medicine, 6e | Medical Education | Health Library (oclc.org)

Objectives

- SOM.MK.II.BPM2.1.ER.1.PHYS.1065 Explain the role of the human male in reproduction and list the physiological functions of human male reproductive tract.

- SOM.MK.II.BPM2.1.ER.1.PHYS.1066 Describe the role of Sertoli cells, Leydig cells and the basement membrane in spermatogenesis.

- SOM.MK.II.BPM2.1.ER.1.PHYS.1067. Explain the age-related changes in the male hypothalamo-pituitary-gonadal axis that lead to puberty, reproductive maturity, and reproductive senescence (andropause).

Objectives

- SOM.MK.II.BPM2.1.ER.1.PHYS.1068. Describe each of these hormones GnRH , FSH, LH and their mechanism of action
- SOM.MK.II.BPM2.1.ER.1.PHYS.1069. List the major target organs for testosterone and other androgens.
- SOM.MK.II.BPM2.1.ER.1.PHYS.1070 Explain the biosynthesis, mechanism of transport within the blood, metabolism and elimination of testosterone and related androgens.

Objectives

- SOM.MK.II.BPM2.1.ER.1.PHYS.1071 Describe the neural, vascular, and endocrine components of the erection and ejaculation response.
- SOM.MK.II.BPM2.1.ER.1.PHYS.1072 Identify the causes and consequences of over secretion and under secretion of testosterone for a) prepubertal and b) postpubescent males c) infertility
- SOM.MK.II.BPM2.1.ER.1.PHYS.1073 Discuss the role of hypothalamic-gonadal axis in the regulation of hormone synthesis.

Role of the human male

- The male human has two major roles with respect to the physiology of reproduction
    - production of sperm (Spermatogenesis)
    - transportation of sperm for fertilization

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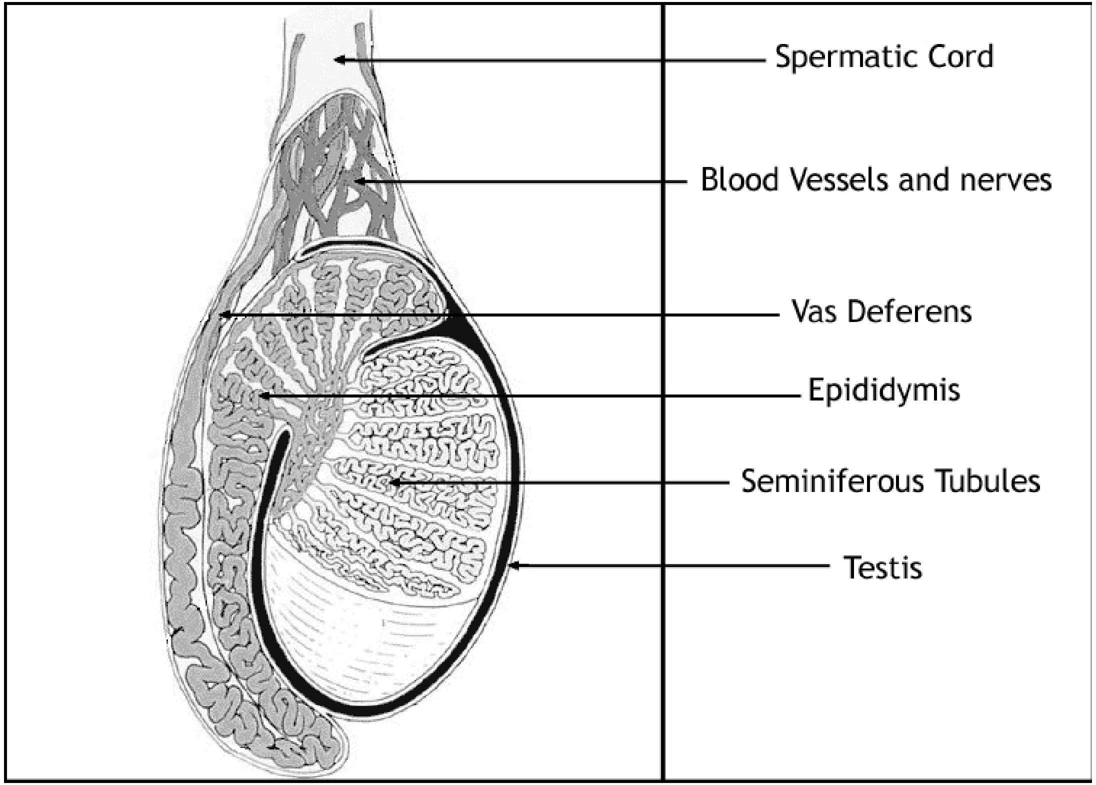
Sagittal Section of the testis



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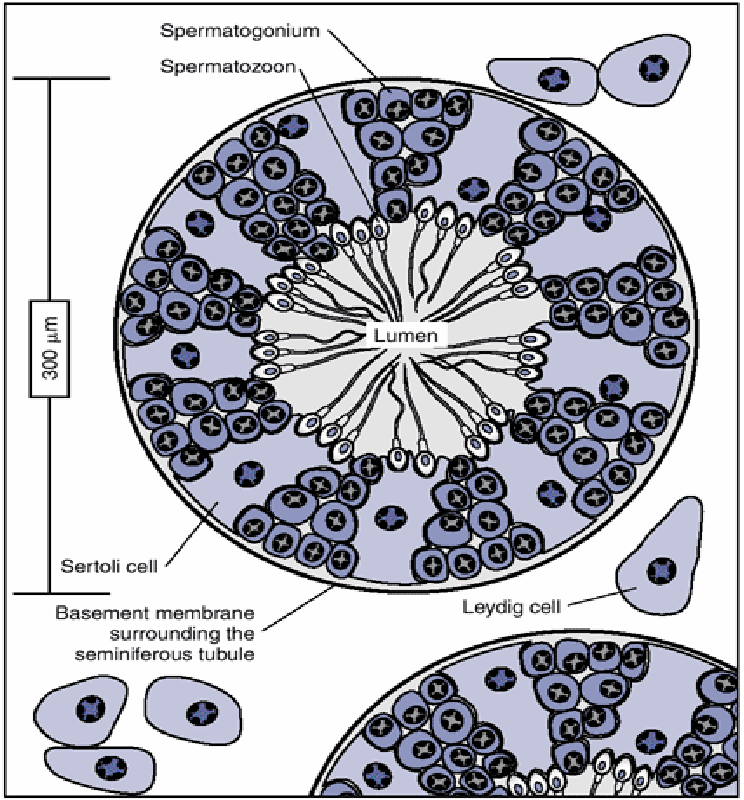
Anatomy - testes



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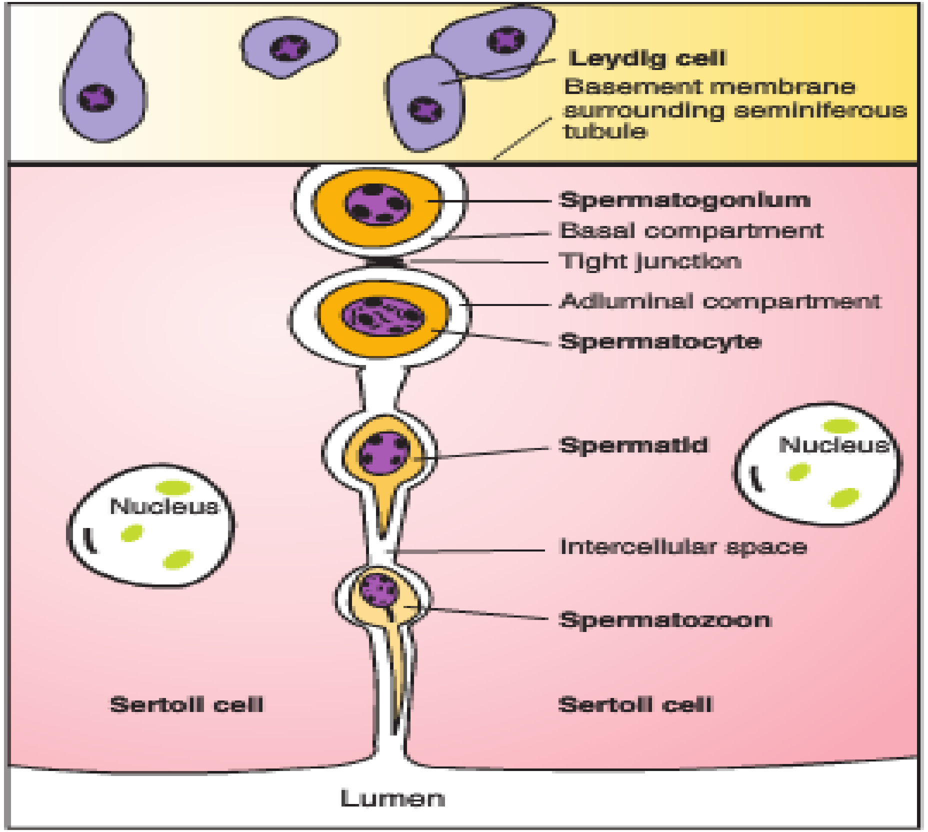
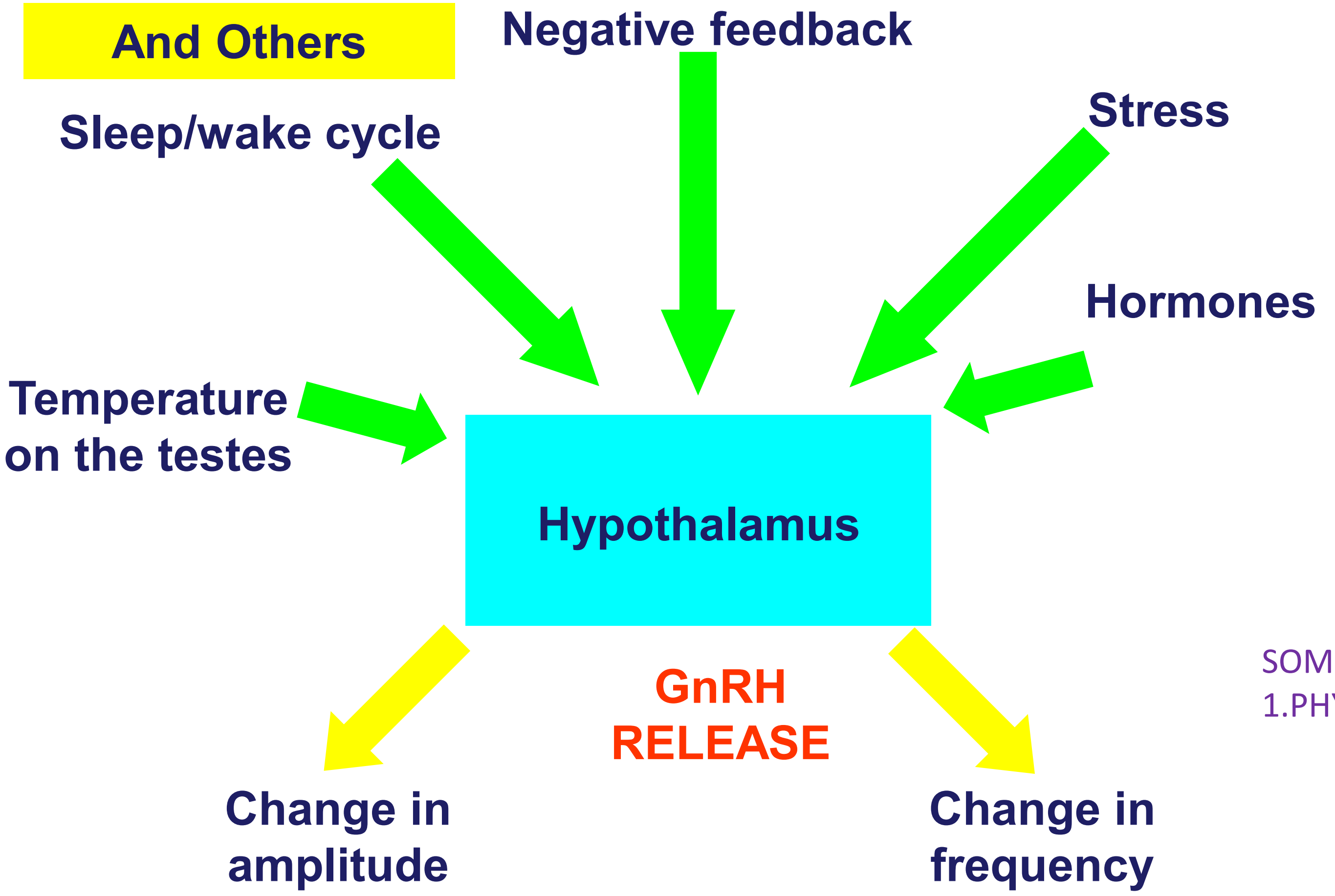
Anatomy - testes



- Sertoli cells are connected by tight junctions, which divide the intercellular space into:
  - basal compartment
  - adluminal compartment.
- Spermatogonia are located in the basal compartment and maturing sperm in the adluminal compartment.
- Spermatocytes are formed from the spermatogonia

Gonadotropin Releasing Hormone (GnRH)



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Hypothalamic pituitary axis

Hypothalamus

↓ **GnRH**

Anterior Pituitary

↓ **LH and FSH**

Testes

↓

Testosterone

Gonadotropin Releasing Hormone (GnRH)

- 10 AA peptide
- Released in median eminence of hypothalamus
- Secreted for a few minutes every 1-3 hours
    - pulsatile

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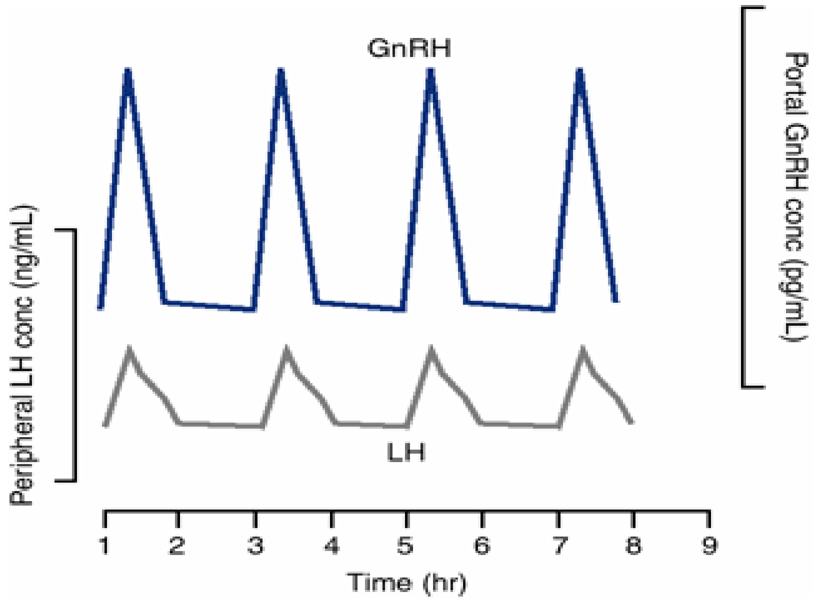
GnRH secretion



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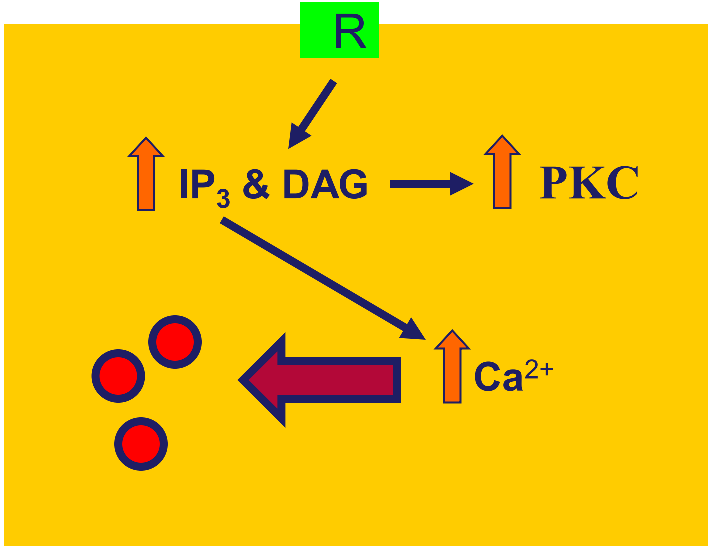
Gonadotroph – anterior pituitary

GnRH

Vesicles containing LH/FSH



Release of LH/FSH via exocytosis

Also ↑ LH and FSH synthesis

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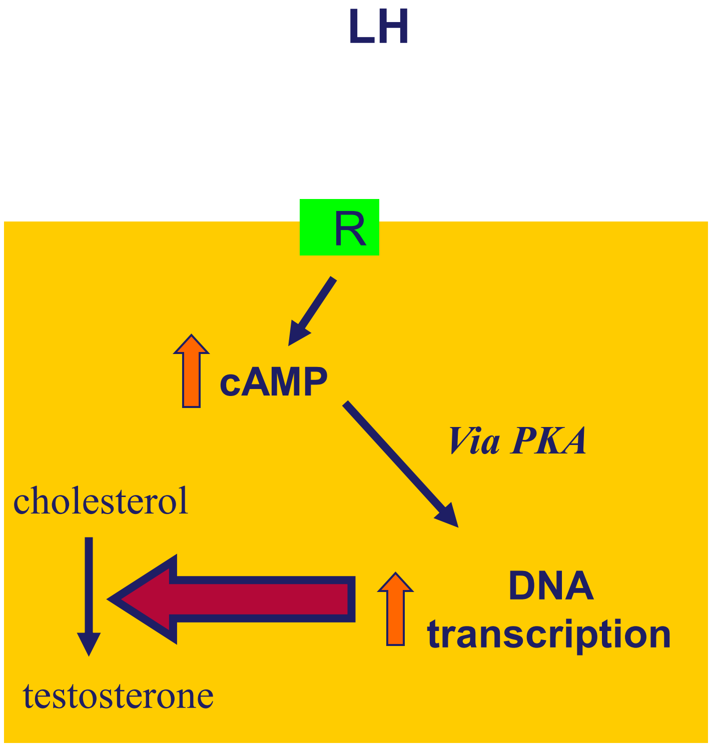
Leydig Cell - testes

Increased testosterone production



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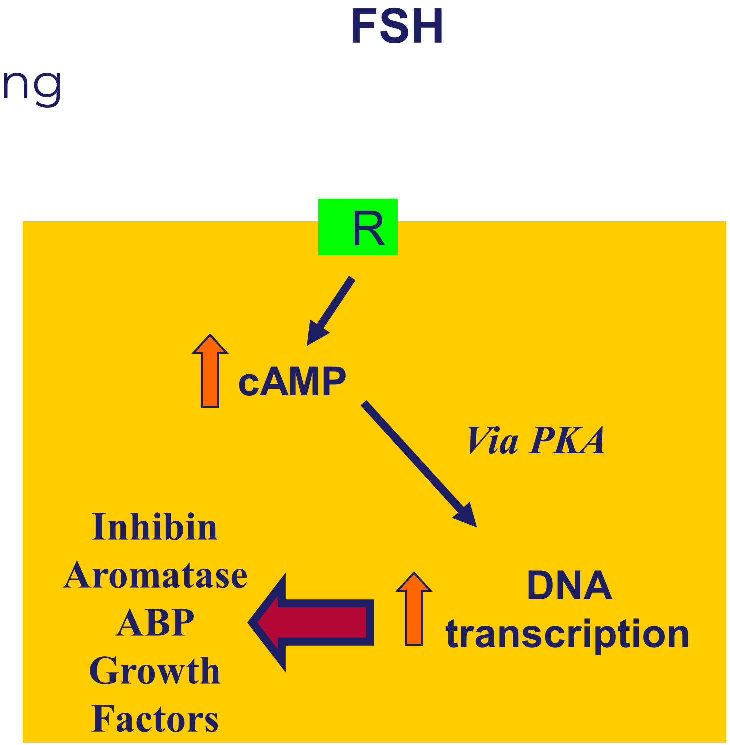
Sertoli Cell - testes

ABP-Androgen binding protein



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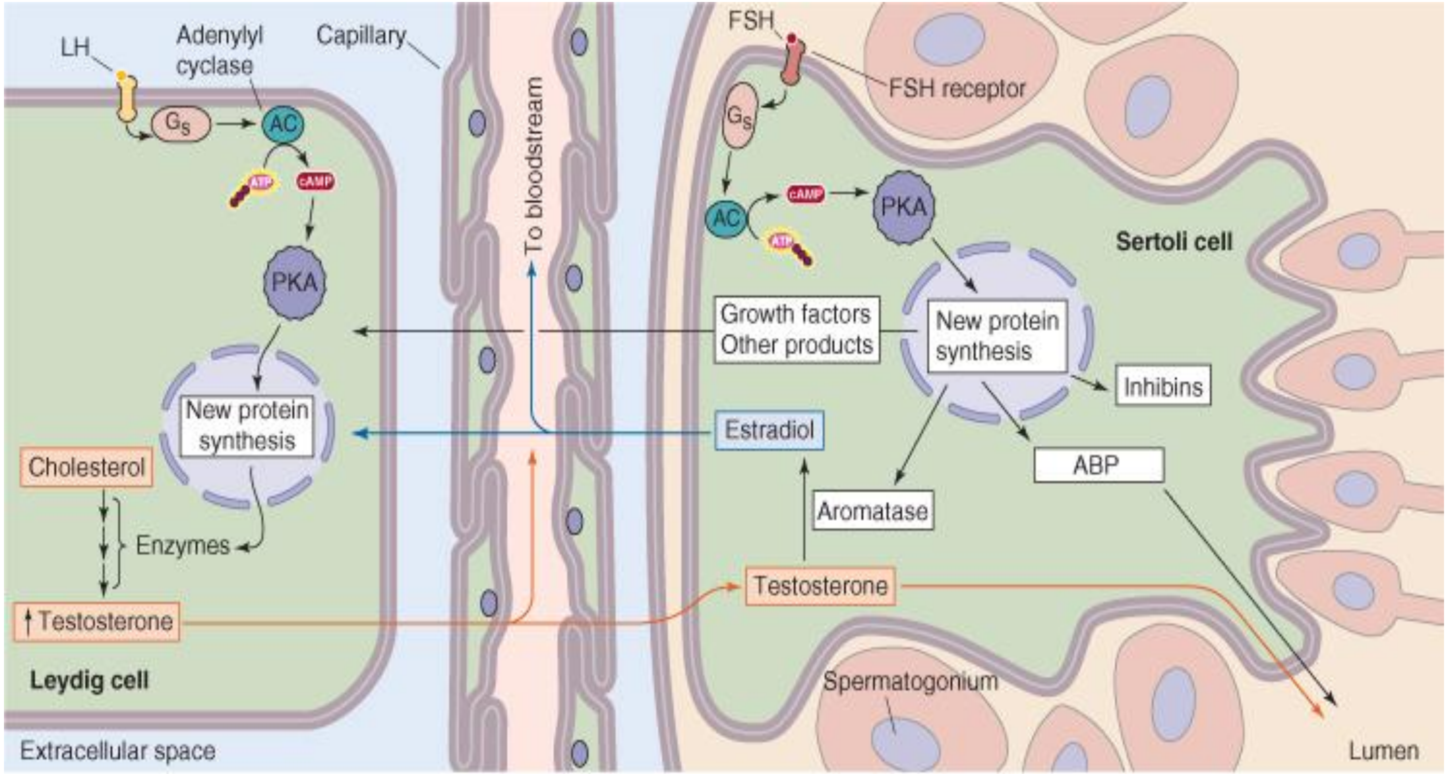
Testes

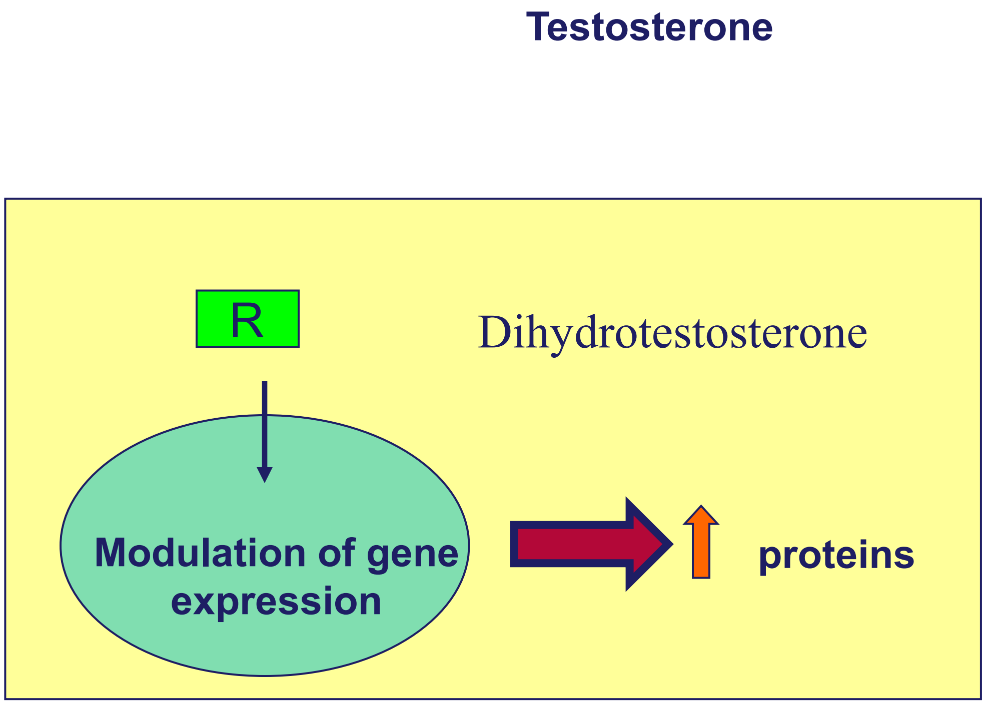


LH
Adenylyl cyclase
Capillary
Gs
AC
ATP
cAMP
PKA
New protein synthesis
Cholesterol
Enzymes
↑Testosterone
Leydig cell
Extracellular space
To bloodstream
FSH
FSH receptor
Gs
AC
cAMP
ATP
PKA
Sertoli cell
Growth factors
Other products
New protein synthesis
Inhibins
Estradiol
ABP
Aromatase
Testosterone
Spermatogonium
Lumen

At the target tissues

Target tissues

- male internal genitalia
- male external genitalia
- bones
- muscles



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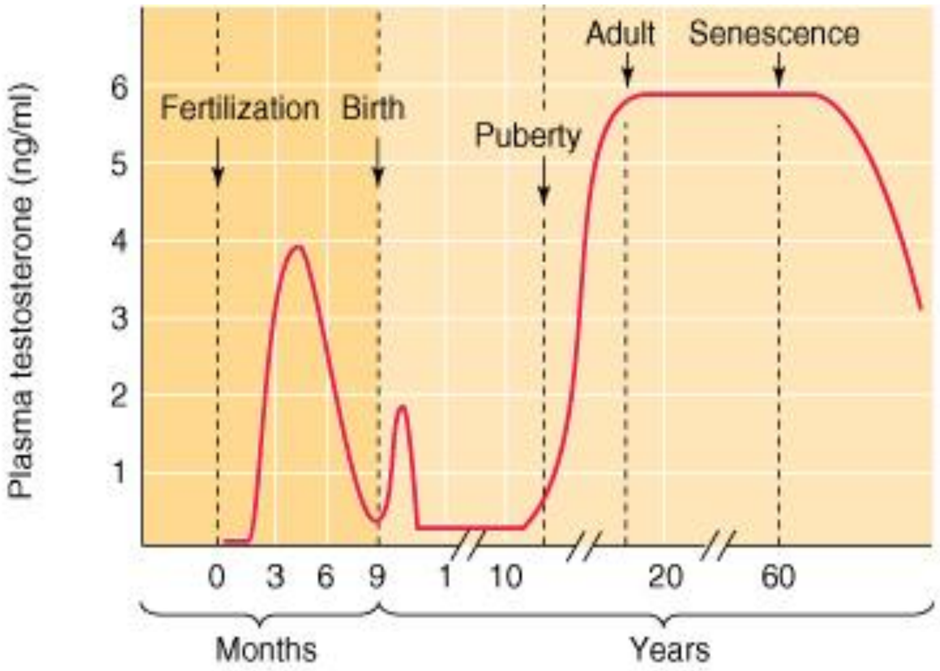
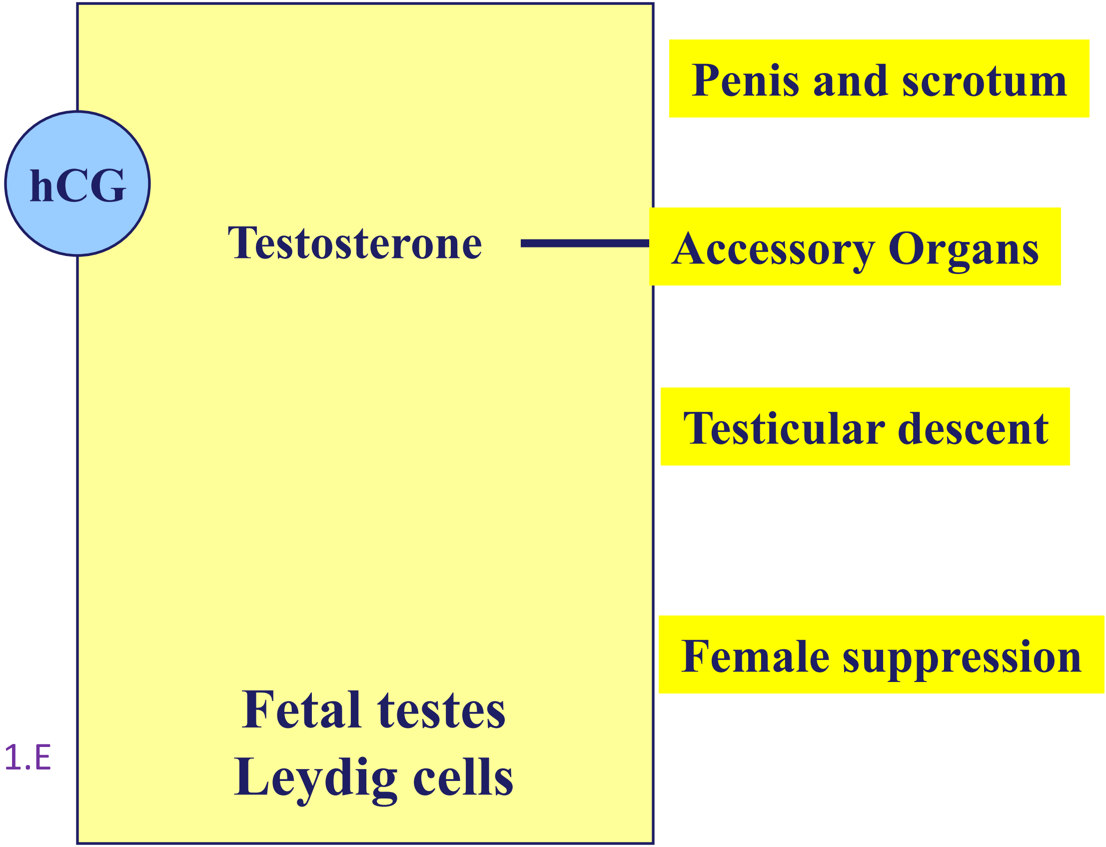


Physiological functions of testosterone

Effects of Testosterone in the fetus



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R.1.PHYS.1069

Spermatogenesis

Already covered in histology

Actions of testosterone - adults

Testosterone

- ↑size of reproductive organs
- promotes spermatogenesis
- sex drive
- development of 2^0 sexual characteristics

- promotes long bone fusion
- ↑BMR
- ↑ in RBC's mass

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2^0 sexual characteristics - adult males

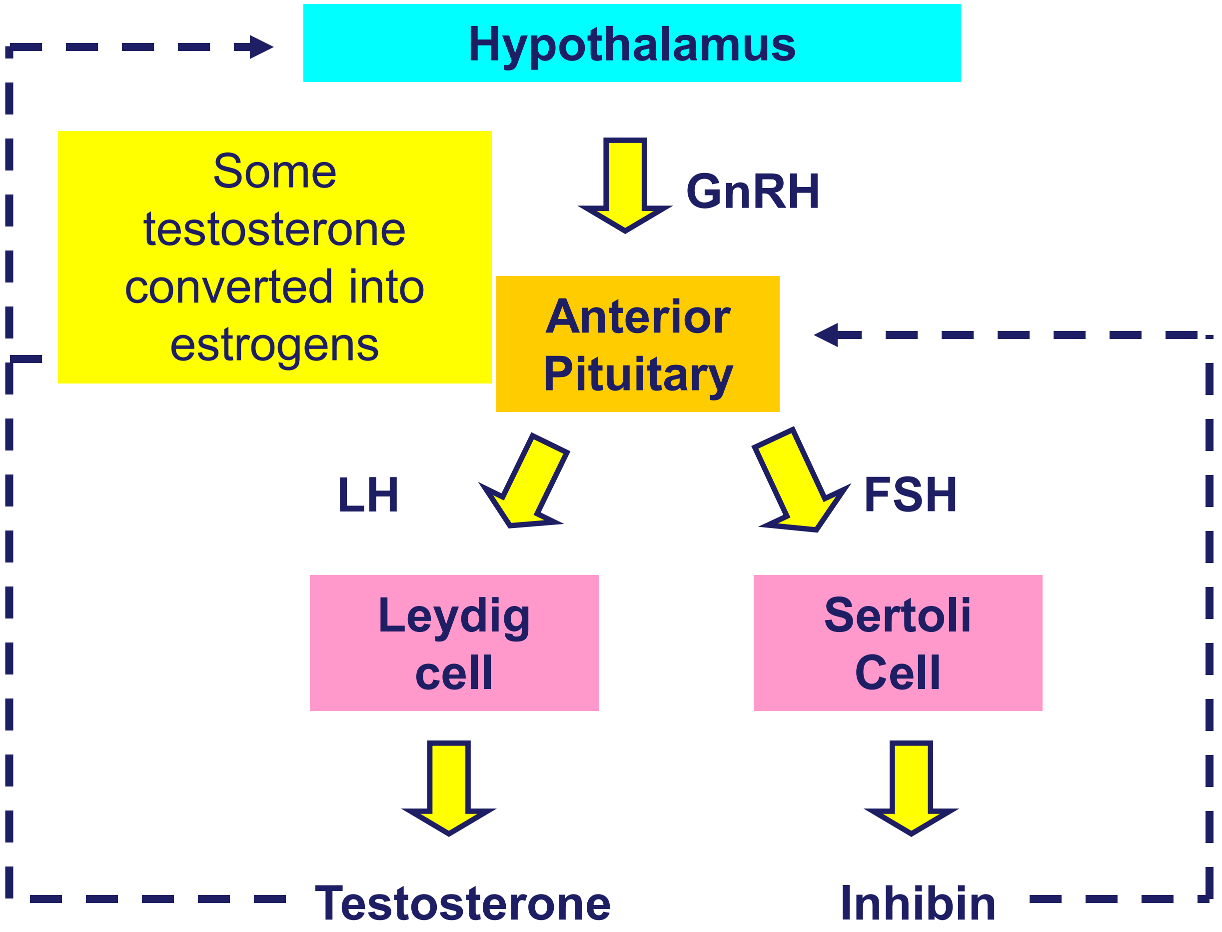
Testosterone

- Effects voice
- thickens skin &
- ↑ sebaceous secretions leading to acne
- male pattern of baldness
- ↑ body hair on face, chest, pubis, linea alba & other portions

- deposition of protein
- ↑ muscle mass
- ↑ strength & size of bones

Testosterone (cont)

- Circulates bound (~97%) to
  - Sex-hormone binding globulin
  - Albumin
- Free testosterone is converted into a more active form dihydrotestosterone (by 5 α reductase) which is more potent than testosterone
- It is metabolized in the liver & excreted by the kidneys



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Formation of the ejaculate

Sperm storage

Epididymis
vas deferens

10%

Seminal vesicles

Sperm, fructose, prostaglandins, clotting precursors

60%

Prostate gland

+ alkalization, semen clotters
And unclotters

30%

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Formation of the ejaculate

The male sex act

The male must have an erection and ejaculate in order to complete the sex act.

Mechanism of erection

PSYCHOLOGICAL AND/OR PHYSICAL STIMULI

↑ Parasympathetic Activity

↓ Sympathetic Activity

Via NO and/or ACh

Vasodilation of the corpora cavernosa and corpus spongiosum

↓

Erection

NO – nitric oxide

Erectile dysfunction or Impotence

Is the inability to have complete erection or have a brief one.

↑ **Parasympathetic Activity**

Via NO &/or Ach they ↑ levels of cGMP by inhibiting its degradation

works

- **Viagra**
- **Cialis**
- **Levitra**

Vasodilation

⇓

Erection

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Ejaculation

Emission

- Genital duct / accessory organ contraction (sympathetic stimulation)
    - Vas deferens, prostate, seminal vesicles
    - Mix with mucus in urethra producing semen

Expulsion

- Semen expelled by rhythmic skeletal muscle contraction at base of penis

Medical problems associated with the male reproductive organs

Hypogonadism

- Defective testosterone secretion in fetus
  - Lead to development of female external genitalia.
- Loss of testes pre-puberty
  - Maintain infantile sex organs and failure to develop male secondary sex characteristics later in life.
  - Bones less strong but longer
  - Become taller than normal men (eunuchs)

Hypogonadism

- Loss of testes post-puberty
  - Slight decrease in size of external and internal genitalia
  - Slight increase in voice pitch
  - Loss of bone and protein build up
  - Slight loss of sexual desire
  - Less easy for erection
  - Ejaculation rarely occurs

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Hypogonadotropic hypogonadism

- The most common congenital form is **Kallmann syndrome**, which results from **decreased or absent GnRH secretion**
- Associated with **anosmia or hyposmia**
- Patients do not undergo pubertal development and have eunuchoidal features.
- Plasma **LH, FSH, and testosterone levels are low**, and the testes are immature and have no sperm.
- Treatment: intermittent treatment with GnRH can produce sexual maturation and full spermatogenesis.

Hypergonadotropic hypogonadism

- Klinefelter syndrome
  - The typical karyotype is 47 XXY.
  - The extra X chromosome interferes with development of the seminiferous tubules and Leydig cell proliferation.
  - Examination findings include gynecomastia, cryptorchidism, testicular **volume is reduced >75%**, and the ejaculates contain few, if any, spermatozoa (azoospermic).
  - **Low plasma testosterone and high gonadotropin level**

Infertility

- Infertility in a couple is defined as the inability to achieve conception despite one year of regular, unprotected intercourse.

- Sperm count
  - Normal sperm count limit is: More than 15 million/mL

- Sperm morphology and motility
  - Abnormal shape of sperm
  - Lack of motility

References

- Medical Physiology – Principles for Clinical medicine – Rodney A Rhoades & David A. Bell - 6th edition
- Endocrinology – Boron & Burlap
- Guyton & Hall – 11th Edition